

AGRI-NEXUS

Integrated AI & IoT Platform for Early Threat Detection
and Post-Harvest Produce Management

Project Domain	Agriculture • Artificial Intelligence • IoT
Innovation Focus	Early threat detection + post-harvest management
Target Users	Farmers, agricultural advisors, buyers and supply-chain stakeholders
Hackathon	Smart India Hackathon 2026
Team / Identity	NEBULA

Healthier Crops → Less Loss → Better Income → Food Security

1. Abstract

AGRI-NEXUS is an integrated Artificial Intelligence (AI) and Internet of Things (IoT) platform designed to support farmers across two connected stages of agricultural production: early threat detection in the field and post-harvest produce management. The platform combines crop images, environmental sensor data and location information to identify disease and pest risks at an early stage.

The system processes field data using computer vision and machine-learning models, generates risk information, stores and synchronizes data through cloud services, and presents actionable information through a mobile/web interface. After harvesting, IoT-based monitoring can track storage conditions such as temperature and humidity, helping reduce avoidable quality deterioration and post-harvest losses.

The overall objective is to move agricultural decision-making from reactive treatment toward data-driven prevention, while supporting better produce handling, traceability and market connectivity.

2. Problem Statement

Farmers can face delayed identification of crop diseases and pests, changing weather and soil conditions, limited access to timely crop-specific guidance, and losses after harvest because of unsuitable storage or handling conditions.

Existing processes may also involve manual monitoring and fragmented information. A practical digital solution therefore needs to connect field sensing, crop-image analysis, risk prediction, farmer alerts and post-harvest monitoring in one workflow.

3. Key Problems Addressed

1. Late Disease & Pest Detection: Crop threats may become visible only after significant damage has occurred, reducing the opportunity for early intervention.

2. Unpredictable Field Conditions: Changes in temperature, humidity, rainfall and soil moisture can influence crop stress and disease/pest risk.

3. Limited Expert Guidance: Farmers may not receive timely, field-specific recommendations in a convenient form.

4. Post-Harvest Losses: Poor storage conditions, delayed handling and insufficient quality monitoring can reduce produce quality and marketable quantity.

5. Fragmented Market Information: Farmers can benefit from better visibility into buyers, markets and produce movement.

4. Proposed Solution – AGRI-NEXUS

AGRI-NEXUS provides a unified digital pipeline that collects field data, detects potential crop threats, predicts risk, delivers localized recommendations and continues monitoring after harvest.

Smart Field Monitoring: IoT sensors and crop images capture soil, weather and crop-condition information.

AI-Powered Threat Detection: Computer vision and ML models analyze images and sensor information for disease/pest indicators and risk levels.

Risk & Location Mapping: GPS/GIS information is used to identify affected or high-risk locations and support field-level prioritization.

Post-Harvest Management: Temperature, humidity and produce-quality information can be monitored to support better storage and handling.

Farmer Alerts & Guidance: Mobile/web interfaces provide alerts, recommendations and reports, with scope for local-language support.

Supply Chain & Market Linkage: The platform can be extended to connect produce movement, buyers and market information.

5. Objectives

- Detect crop diseases and pests at an early stage using AI-based image analysis.
- Combine crop images with environmental IoT data for improved risk assessment.
- Provide actionable, crop-specific recommendations to farmers.
- Visualize risk and affected areas using GIS/location data.
- Monitor post-harvest storage conditions and reduce preventable losses.
- Create a scalable architecture that can support multiple crops and regions.
- Enable future integration with supply-chain and market systems.

6. Methodology

- Step 1 – Data Collection:** Capture crop images through a mobile device, field camera or drone and collect temperature, humidity, soil-moisture, rainfall and GPS information.
- Step 2 – Pre-processing & AI Analysis:** Clean and enhance images, prepare sensor data, identify disease/pest patterns and estimate severity or risk using ML models.
- Step 3 – Risk & GIS Mapping:** Combine predictions with location information to identify hotspots, generate field-level risk maps and prioritize high-risk zones.
- Step 4 – Cloud & Data Management:** Store, synchronize and process data using a cloud database/backend so that authorized users can access updated information.
- Step 5 – Web Dashboard & Analytics:** Display maps, alerts, trends, reports and post-harvest information through a web dashboard.
- Step 6 – Farmer Alert & Recommendation:** Deliver localized alerts, crop-specific IPM/treatment guidance and storage recommendations through a farmer-facing mobile/web interface.

7. System Architecture

The proposed architecture follows a modular pipeline:

Layer	Main Components	Purpose
Data Sources	Mobile/camera/drone images, IoT sensors, GPS	Capture crop, environment and location data
Edge / Pre-processing	Image enhancement, cleaning, data fusion	Prepare reliable input for analysis
AI/ML Engine	Computer vision, disease/pest detection, risk prediction	Classify threat and risk insights
Cloud Platform	Firebase/PostgreSQL, APIs, data synchronization	Store and manage data
Dashboard	GIS maps, alerts, reports, analytics	Visualize information and trends
Farmer Interface	Mobile/web, local-language alerts, recommendations	Deliver actionable guidance
Post-Harvest Layer	Temperature/humidity monitoring, quality and inventory tracking	Support storage and produce management

8. Technical Stack

Hardware: ESP32/Arduino-class IoT controller, temperature and humidity sensors, soil-moisture sensor, optional rainfall sensor, camera and GPS.

AI/ML: Python with TensorFlow or PyTorch; OpenCV for image processing; computer-vision models for disease/pest detection; ML models for risk prediction.

Backend & Cloud: Firebase and/or AWS/ThingSpeak; Node.js or Python APIs; PostgreSQL or cloud database depending on deployment requirements.

Frontend: Flutter/React Native for mobile applications and React.js or similar web technology for the dashboard.

Mapping: GIS tools and map APIs for hotspot visualization, field-level risk maps and location-based analytics.

Communication: MQTT/HTTP for transmitting IoT data from field devices to the backend.

Deployment: Docker can be used for repeatable backend/model deployment.

9. Technical Flow – How AGRI-NEXUS Works

Field sensors and cameras collect crop, soil and environmental data → data is transmitted through an IoT gateway/API → cloud services clean and store the information → AI models analyze crop images and sensor patterns → risk and location information is generated → alerts and recommendations are delivered through the mobile/web interface → post-harvest storage data is monitored → dashboards provide reports and analytics.

This flow creates a continuous feedback loop in which new observations can improve monitoring and support progressively better decision-making.

10. Implementation Plan

Phase 1 – Prototype: Select one crop and collect representative image and sensor datasets.

Phase 2 – AI Model: Train/test disease and pest detection and risk-classification models.

Phase 3 – IoT Integration: Connect sensors and field devices to the backend using MQTT/HTTP.

Phase 4 – Application: Build farmer mobile/web screens for alerts, recommendations and data visualization.

Phase 5 – Post-Harvest Module: Add storage-condition sensing and produce monitoring.

Phase 6 – Validation & Scaling: Test under different field conditions and extend the architecture to additional crops and regions.

11. Expected Outcomes and Impact

- Earlier identification of disease and pest threats.
- More data-driven crop-management decisions.
- Better targeting of interventions instead of unnecessary blanket treatment.
- Improved monitoring of storage conditions after harvest.
- Reduction of avoidable post-harvest losses.
- Better visibility of crop condition and risk through GIS dashboards.
- Potential improvement in produce quality, market readiness and farmer income.
- A scalable digital framework for multi-crop and multi-region deployment.

12. Innovation

The core innovation is the integration of multiple technologies into one agriculture workflow rather than treating crop diagnosis, risk prediction and post-harvest management as separate systems. AGRI-NEXUS connects AI-based crop analysis, IoT sensing, GIS, cloud data management and farmer-facing recommendations.

The concept can be summarized as: **Detect** → **Predict** → **Preserve** → **Connect**.

13. Future Scope

The initial prototype can focus on one crop and a limited set of diseases/pests for controlled validation. Future versions can expand to multiple crops and regions, add more disease classes, improve predictive analytics, integrate satellite or drone imagery, support multilingual voice assistance, strengthen supply-chain traceability, and connect farmers with buyers and market services.

14. Conclusion

AGRI-NEXUS proposes a practical AI and IoT-based approach for improving agricultural decision-making from the field through the post-harvest stage. By combining early threat detection, environmental monitoring, risk mapping, cloud analytics, farmer alerts and storage-condition monitoring, the platform aims to reduce avoidable losses and support more informed agricultural practices.

The architecture is modular and scalable, allowing the prototype to begin with a focused crop-specific implementation and progressively expand into a broader digital agriculture ecosystem.

Project Identity: NEBULA | **Project:** AGRI-NEXUS | **Event:** Smart India Hackathon 2026